Welcoming remarks

Neonatal AMR genomic surveillance of *Klebsiella pneumoniae* and *Escherichia coli*
KlebGen training course · Africa CDC and ILRI

About 1,300 words, roughly 10 minutes at a relaxed pace. Text in [square brackets] is to be filled in. Timings are approximate; trim sections 4 or 6 if time is short.

1. Welcome (about 1 minute)

Good morning, everyone, and welcome.

My name is [name]. On behalf of [ILRI, Africa CDC and the KlebGen team], it is a real pleasure to have you here in [city] for the next ten days.

Some of you have travelled a long way to be in this room: from [number] countries, from national public health institutes, reference laboratories, hospitals and universities. Thank you for coming. Thank you also to your institutions and supervisors, who let you step away from busy labs and wards for two full weeks. That is not a small thing, and we don't take it for granted.

I also want to thank [funders / Africa CDC Pathogen Genomics Initiative / host institution], and the trainers and support staff, many of whom have been preparing this course for months.

2. Why we are here (about 2 minutes)

Let me start with why this course exists.

Every one of us has seen, or heard about, a newborn baby with a bloodstream infection. Neonatal sepsis remains one of the leading causes of death in the first month of life, and in many hospitals across our continent the organisms behind it are *Klebsiella pneumoniae* and *Escherichia coli*.

What makes this worse is resistance. The antibiotics recommended for empiric treatment of sick newborns are ampicillin with gentamicin, then a third-generation cephalosporin. For *Klebsiella*, ampicillin does nothing on its own, so the first-line regimen depends on gentamicin alone. When the bacterium also carries a gentamicin resistance gene, or an ESBL, or a carbapenemase, the drugs a clinician reaches for first may simply not work. That baby may have very little time to wait for a second attempt.

And these infections do not happen at random. They cluster: in one nursery, around one sink, one incubator, one period. Sometimes the same strain moves from baby to baby. Sometimes a resistance gene moves between strains on a plasmid, which is harder still to see.

So the questions we will keep asking over these ten days are very practical ones:

- What is causing these infections?
- Which antibiotics will still work?
- Are these cases linked?
- If they are, what should the hospital do on Monday morning?

3. Why genomics, and why together (about 1.5 minutes)

Whole-genome sequencing lets us answer those questions with a resolution we never had before. From one genome we can:

- confirm the species,
- find the lineage,
- list the resistance and virulence genes,
- see which plasmid carries them,
- and, by comparing genomes, tell whether infections are part of one outbreak or just happen to look alike.

But a sequencer on its own answers nothing. Between the raw reads and a decision in a neonatal unit there is a lot of work: quality control, assembly, typing, careful interpretation, and clear communication with clinicians and infection prevention teams. That middle part is what this course is about.

We are doing this as a network on purpose. Surveillance is only as strong as its weakest link. A resistant clone does not respect borders, and neither do plasmids. When each of our countries can generate and interpret these data in the same way, our results become comparable. Then we can see regional patterns, share early warnings, and speak with one voice to policymakers. That is the vision behind [KlebGen / the Africa CDC Pathogen Genomics Initiative], and you are a central part of it.

4. What the next ten days look like (about 2 minutes)

Let me give you a quick map of the course. It follows the path of a real genome, from the sequencer to a decision:

- **Day 1:** we get comfortable with the Linux command line and the computing environment, and look at read quality.
- **Day 2:** we assemble genomes, from short reads and also using long reads.
- **Days 3 and 4:** we ask who the organism is and what resistance it carries. We will also compare genome predictions with what the laboratory measured, because a genome is only useful if we know how far to trust it.
- **Days 4 and 5:** we look at plasmids, the mobile pieces of DNA that carry so much of the resistance we worry about.
- **Days 6 and 7:** we build trees, count SNPs and ask the outbreak question. On Day 7 we finish with a role play where you present your findings to an infection prevention team. It is one of the most useful exercises of the course, and also one of the most fun.
- **Days 8 and 9:** we look at capsule types, which matter for future vaccines, and bring everything together into figures and a short brief.
- **Day 10:** the capstone: you put everything into practice on your own.

Throughout, we use real data, from a published neonatal study in Blantyre, Malawi. It has a real cluster in a real neonatal unit, so the questions you answer are the ones that team faced.

5. Your own data (about 1.5 minutes)

One part of this course I'm particularly excited about comes on Day 5.

By then you will have done every step by hand. We then show you the course pipeline, which runs all of those steps in one command, the same way for every genome. And you will not run it on our data: each country will run it on its own genomes.

Each country team will bring its assemblies and a metadata table. One participant per country will submit the job, or a trainer will submit it on your behalf, because the computing resources are limited and we want to use them well. Each country's results stay in their own folder, and they belong to that country team.

So when you go home, you won't just take a certificate. You'll take a working analysis of your own country's isolates, and the skills to repeat it.

6. How we will work together (about 1.5 minutes)

A few practical points before we begin.

- **This is a hands-on course.** Most of your time will be spent typing commands, reading output and discussing what it means. Things will break, and error messages will appear. That is normal; it is how everyone learns this work.
- **Please ask questions,** early and often. If you are stuck for more than a few minutes, put up your hand or tell your neighbour. The trainers are here for exactly that. Your question is very likely someone else's question too.
- **Help each other.** Many of you have skills the rest of the room does not, whether in the lab, at the bedside, in programming or in public health. Some of the best learning this week will happen between participants, not from the front of the room.
- **On computing:** you will each run the steps on a few practice isolates, and the trainers have computed the rest in advance so you can copy those results and move on. If your connection drops or your session times out, don't panic: the course notes explain how to pick up where you left off.
- **On data:** treat the data with care. Some of it comes from real patients, and very soon some of it will come from your own countries.
- **[Logistics:** daily timing, breaks, lunch, Wi-Fi, who to contact, certificate requirements.]

7. Closing (about 30 seconds)

Let me end where I started.

Behind every isolate in our datasets is a newborn baby, and a family hoping that the next antibiotic will work. The skills you build over these ten days can help hospitals find outbreaks sooner, choose treatments more wisely, and give our countries the evidence they need to act.

So I hope you leave tired, but confident. I hope you leave connected to colleagues across the continent whom you can call on long after this course ends. And I hope you leave a little bit excited about what your laboratory can now do.

Welcome, once again. Let's get started.